Part 1 link

- https://drive.google.com/file/d/1O5ct8INURfB8H9MQchAa74l7cmOl1QL9/view?usp=sharing

Edit distance

เทียบความยาวของสองสตริงที่เกิดจากการพิมพ์ผิด

apple	ความแตกต่าง
aple	
a-a ตรง	0
p-p ตรง	0
p-l การแทนที่	1
l-e การแทนที่	1
e-_ การลบ	1
	= 3

apple	ความแตกต่าง
aple	
a-a ตรง	0
p-p ตรง	0
p-_ การลบ	1
l-l ตรง	0
e-e ตรง	0
	= 1

เทียบความยาวของสองสตริงที่เกิดจากการพิมพ์ผิด

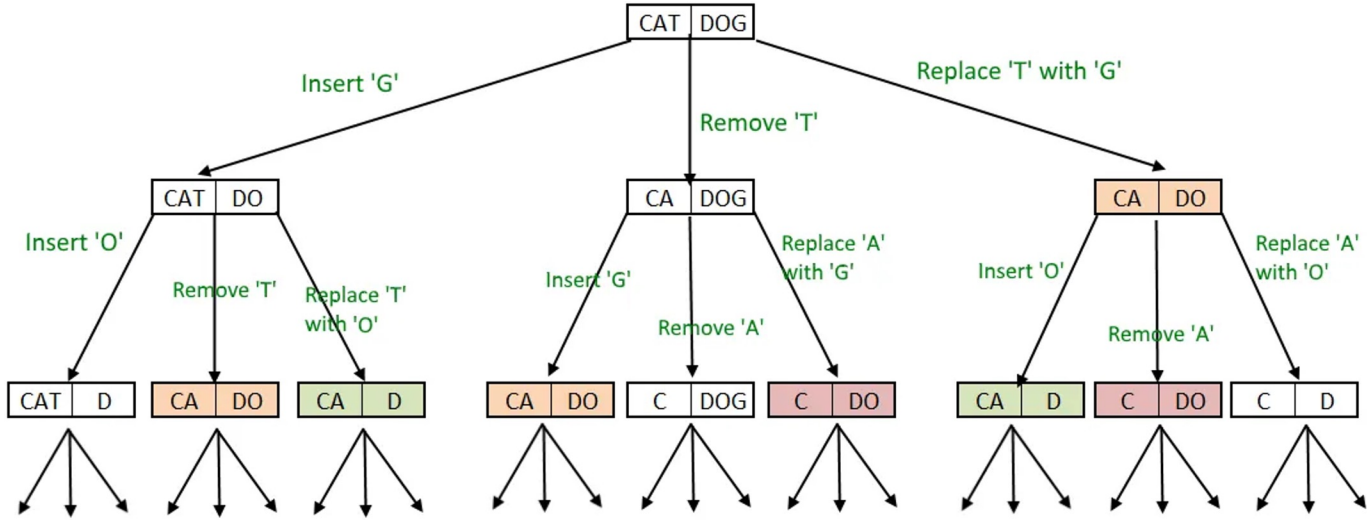
apple	ความแตกต่าง
aple	
a-a ตรง	0
_-p การเติม	0
p-l การแทนที่	1
p-e การ	1
l-_ การลบ	1
e-_ การลบ	1
	=4

apple	ความแตกต่าง
aple	
a-a ตรง	0
p-p ตรง	0
p-l การแทนที่	1
l-e การแทนที่	1
e-_ การลบ	1
	= 3

คำนวณได้หลายวิธีเอาอันที่น้อยที่สุด

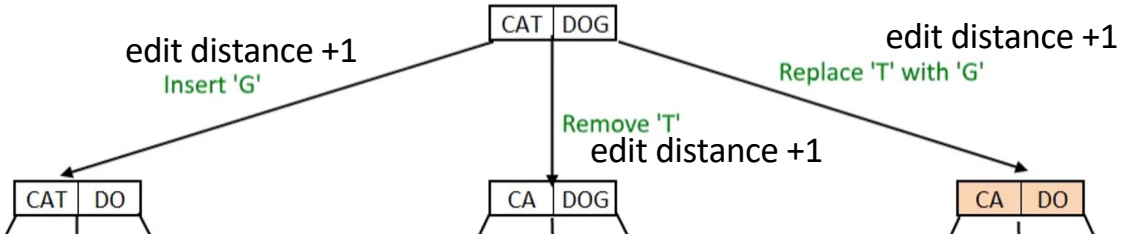
ใช้ recursion

ช่วย

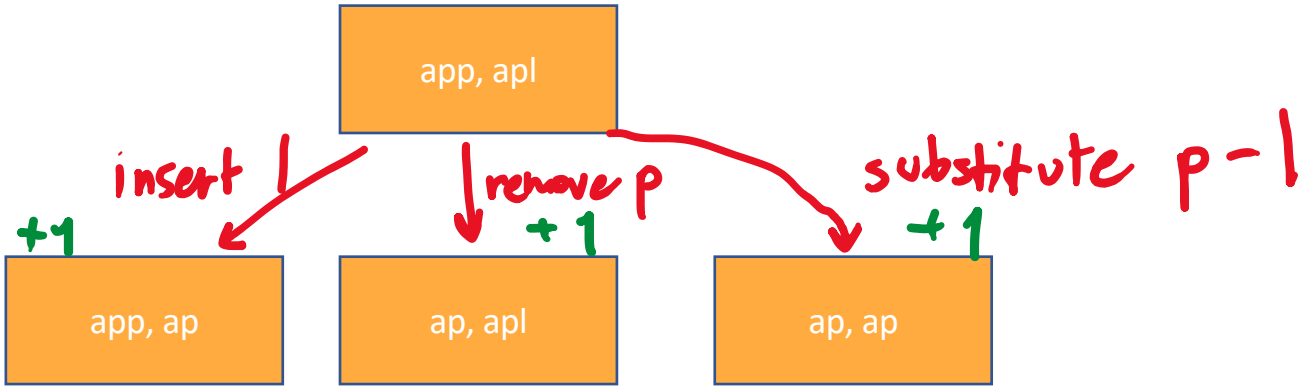


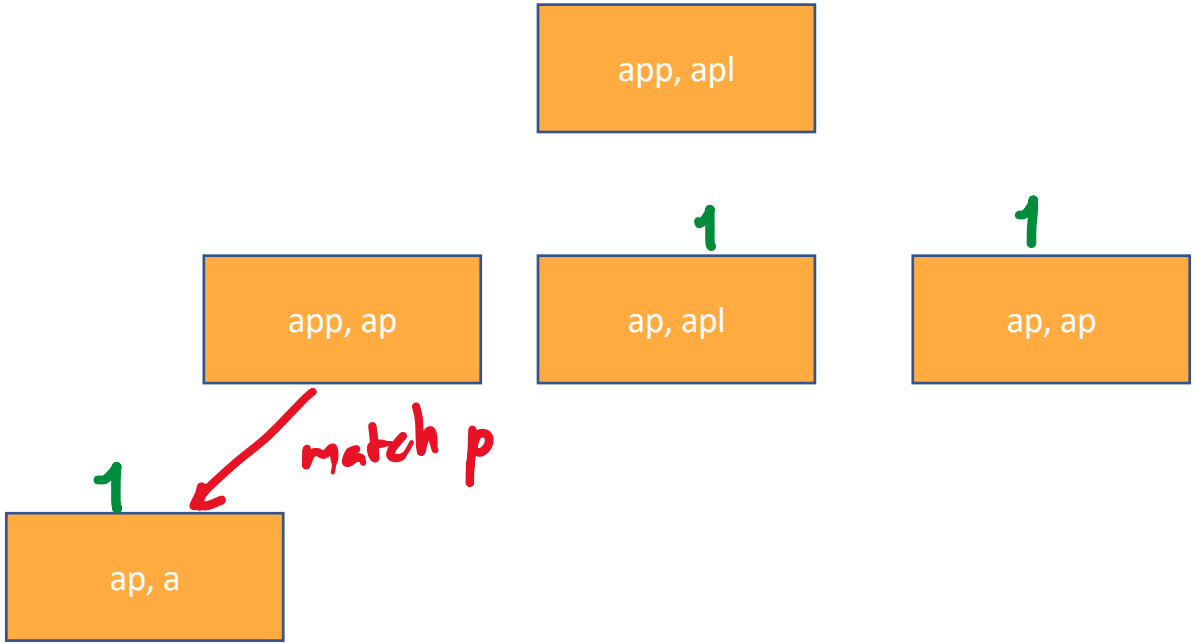
คำนวณได้หลายวิธีเอาอันที่น้อยที่สุด

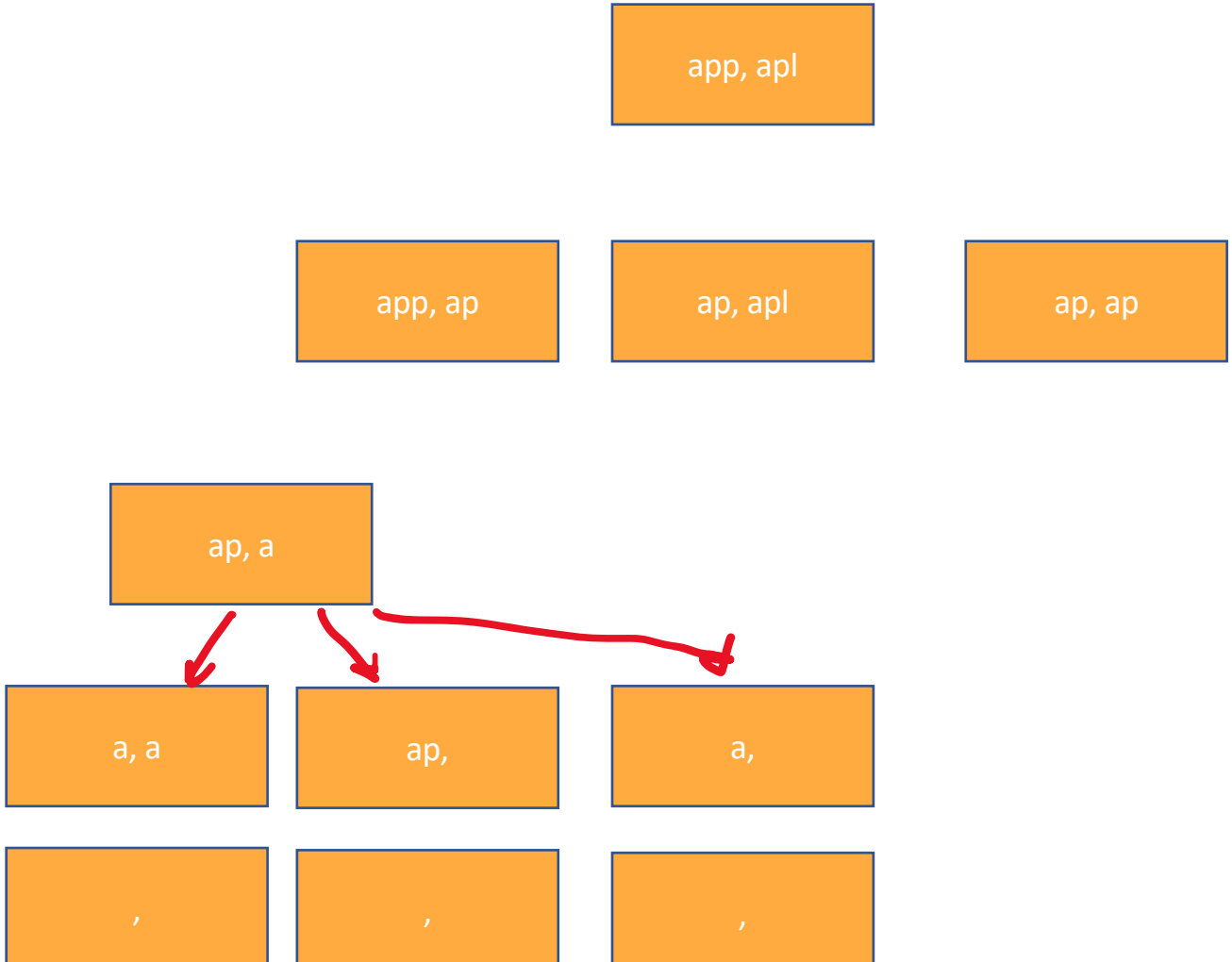
ใช้ recursion ช่วย

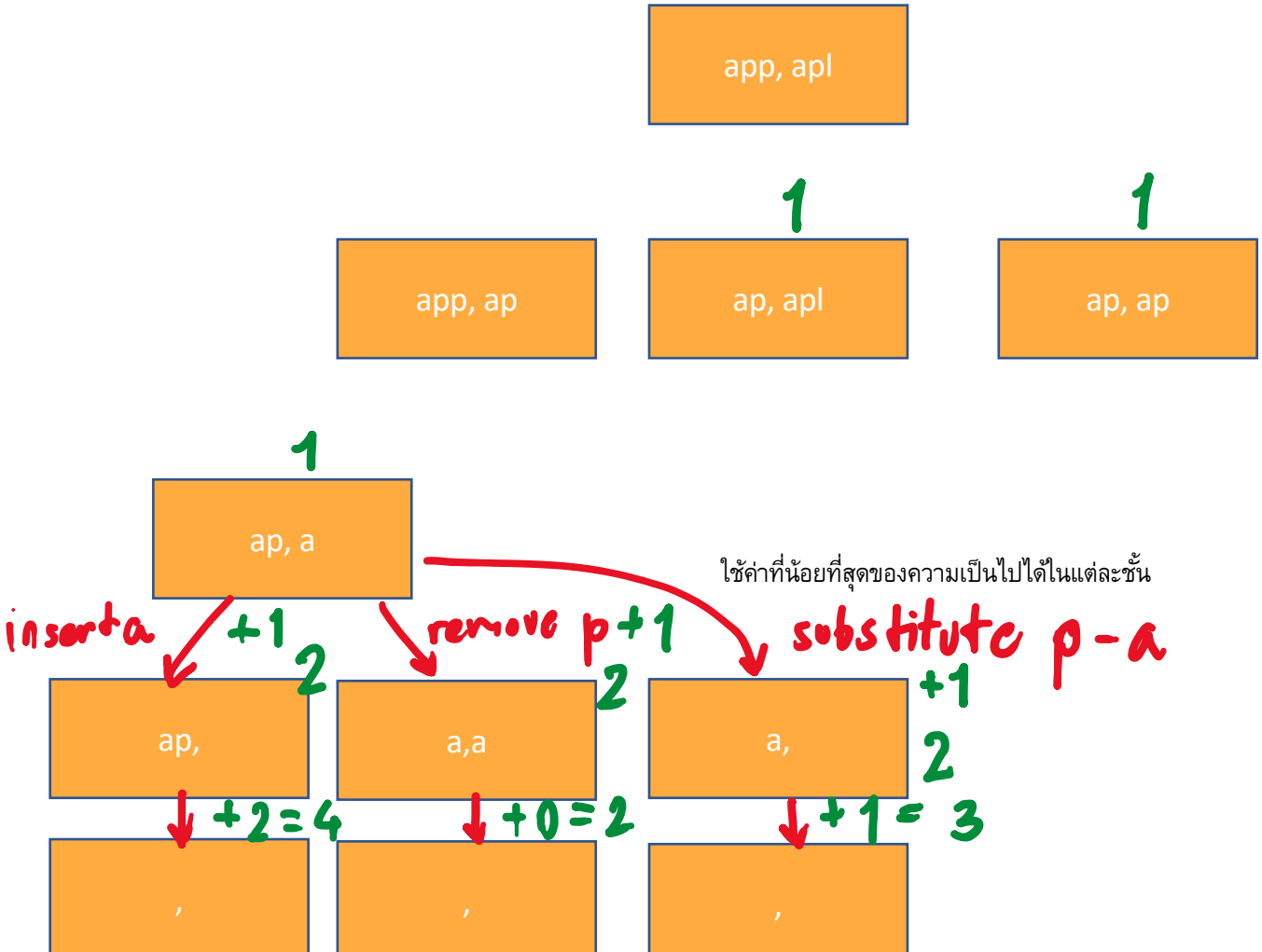


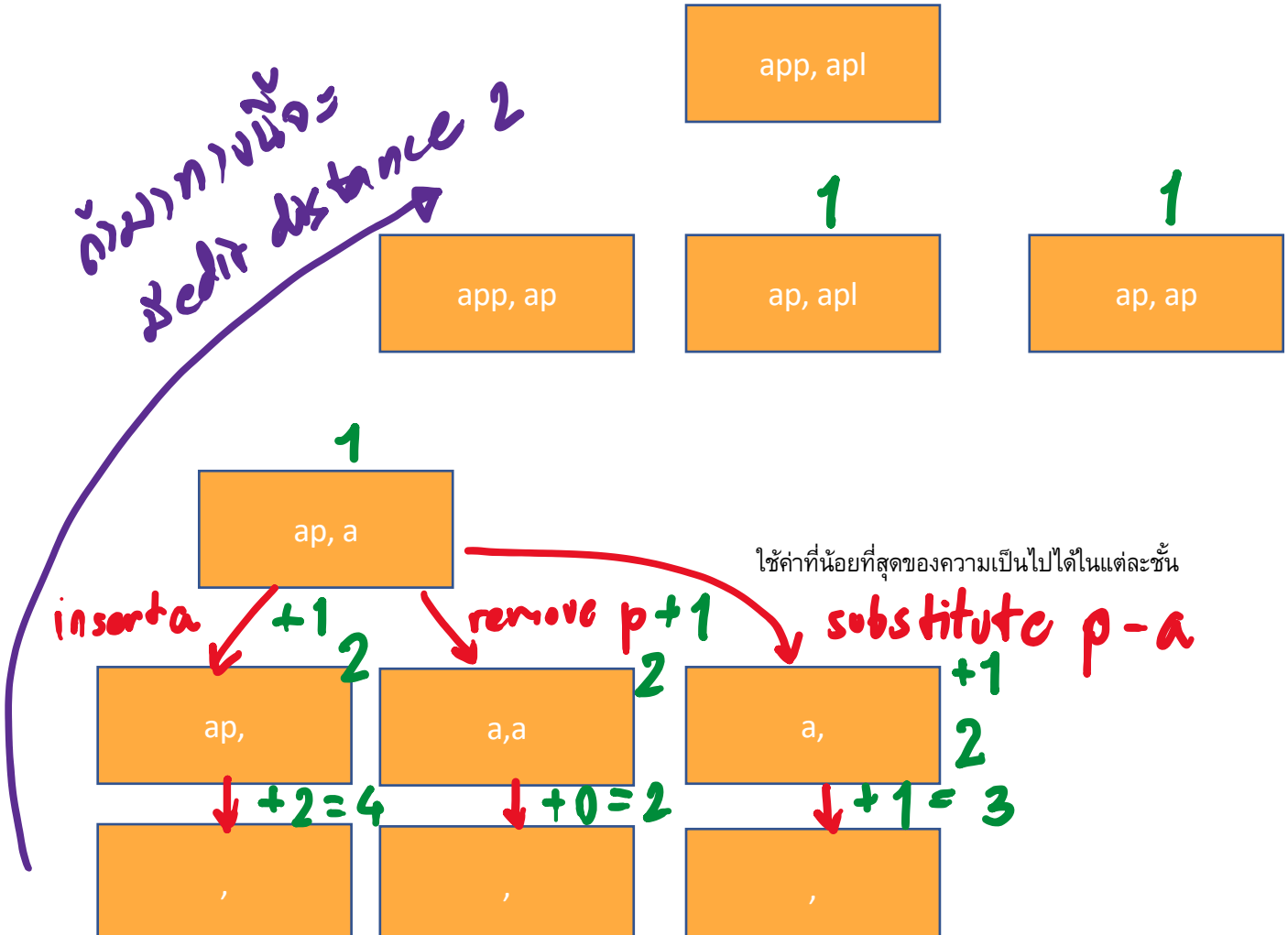
edit distance ของ cat กับ dog เกิดจากedit distanceที่ดีที่สุดจากกรณีที่ล่างลงไป

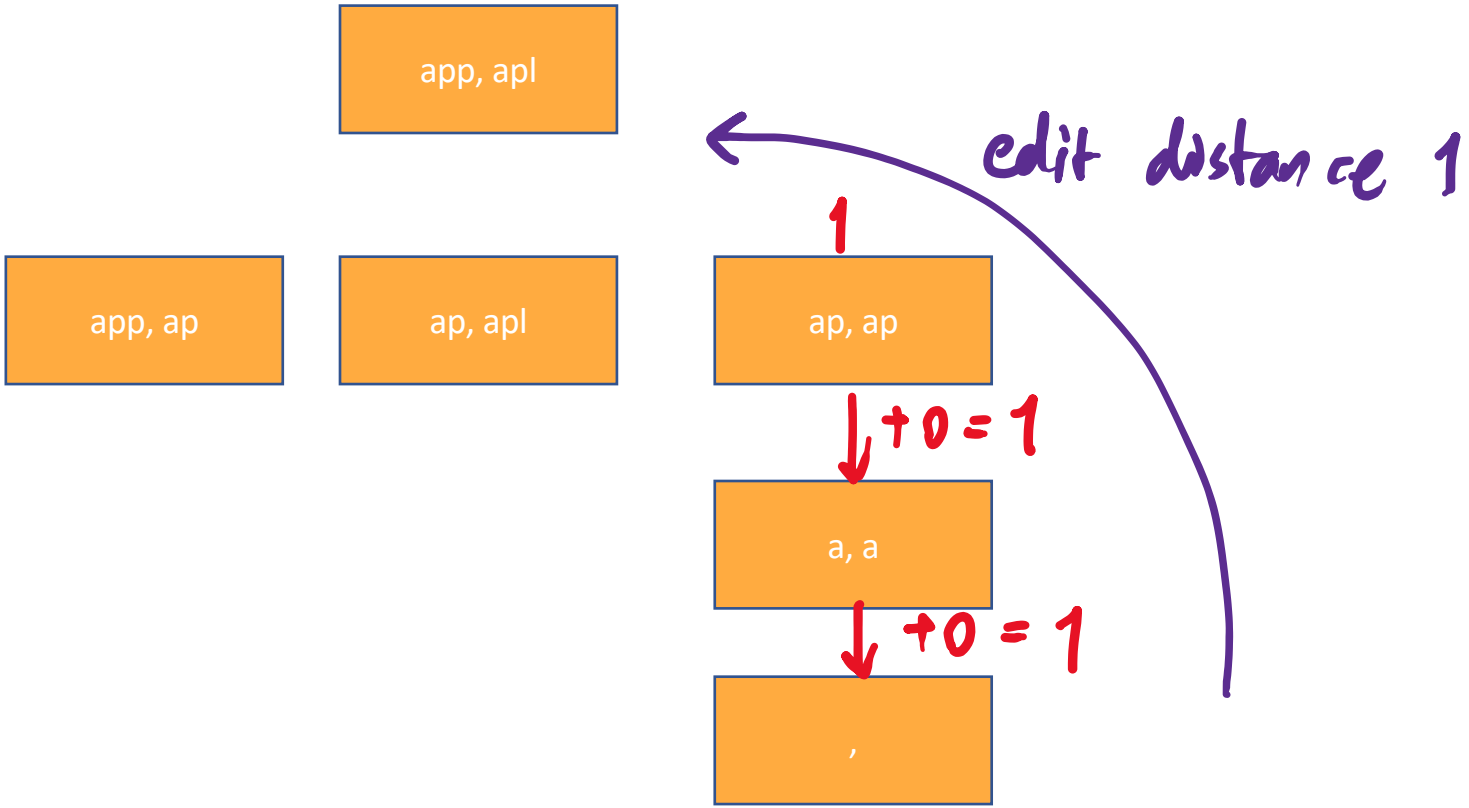

app, apl
insert l
remove p
substitute p-l
+1
+1
+1
app, ap
ap, apl
ap, ap


app, apl
app, ap
1
ap, apl
1
ap, ap
match p
1
ap, a

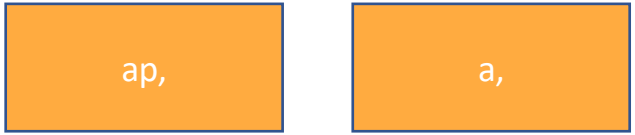

app, apl
app, ap
ap, apl
ap, ap
ap, a
a, a
ap,
a,
,
,
,


app, apl
app, ap
1
ap, apl
1
ap, ap
1
ap, a
ใช้ค่าที่น้อยที่สุดของความเป็นไปได้ในแต่ละชั้น
insert a
+1
remove p+1
substitute p-a
2
ap,
2
a,a
+1
2
a,
+2=4
+0=2
+1=3
,
,
,


app, apl
app, ap
1
ap, apl
1
ap, ap
1
ap, a
ใช้ค่าที่น้อยที่สุดของความเป็นไปได้ในแต่ละชั้น
insert a
+1
remove p+1
substitute p-a
2
ap,
2
a,a
+1
2
a,
+2=4
+0=2
+1=3
,
,
,
ถ้ามาทางนี้จะมี edit distance 2

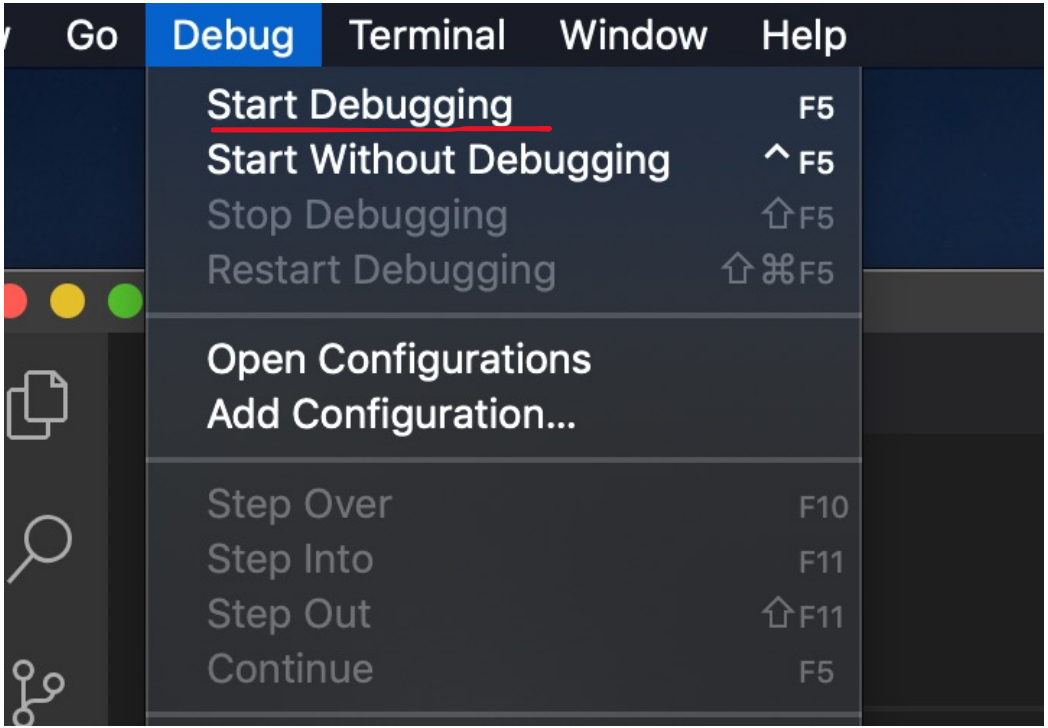

app, apl
edit distance 1
1
app, ap
ap, apl
ap, ap
+0 = 1
a, a
+0 = 1
,

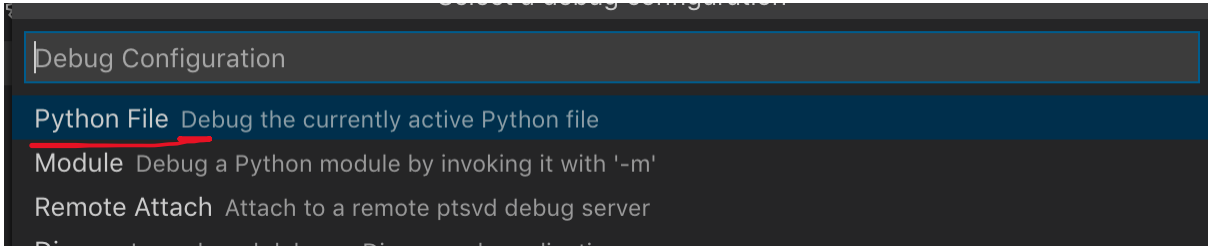
Base case?



- ไม่เหลือตัวอักษรฝั่งซ้าย
- ไม่เหลือตัวอักษรฝั่งขวา

Using VSCode debugger


Go
Debug
Terminal
Window
Help
Start Debugging F5
Start Without Debugging ^F5
Stop Debugging ⇧F5
Restart Debugging ⇧⌘F5
Open Configurations
Add Configuration...
Step Over F10
Step Into F11
Step Out ⇧F11
Continue F5


Debug Configuration
Python File Debug the currently active Python file
Module Debug a Python module by invoking it with '-m'
Remote Attach Attach to a remote ptsvd debug server

test2.py — python2

DEBUG No Conf

VARIABLES

ส่วนแสดงตัวแปร

WATCH

CALL STACK

BREAKPOINTS

- [ ] Raised Exceptions
- [x] Uncaught Exceptions

test2.py

test2.py > ...

```python
a = input()
b = 12
if a > 10:
    a -= 10
    b = b+a
    if a > 10:
        b = b+a
elif: a == 10:
    print(10)
else:
    print("a is not big enough.")

print(a+b)
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL 3: Python Debug Console

```
    main(ptvsdArgs)
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/
__main__.py", line 432, in main
    run()
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/
__main__.py", line 316, in run_file
    runpy.run_path(target, run_name='__main__')
  File "/anaconda3/lib/python3.7/runpy.py", line 261, in run_path
    code, fname = _get_code_from_file(run_name, path_name)
  File "/anaconda3/lib/python3.7/runpy.py", line 236, in _get_code_from_file
    code = compile(f.read(), fname, 'exec')
  File "/Users/ekapolc/Documents/python2/test2.py", line 8
    elif: a == 10:
        ^
SyntaxError: invalid syntax
bash-3.2$
```

ส่วนแสดงการทำงาน

test2.py — python2

DEBUG No Conf กดตรงนี้เพื่อเริ่ม

test2.py

VARIABLES

ส่วนแสดงตัวแปร

WATCH

CALL STACK

BREAKPOINTS

- [ ] Raised Exceptions
- [x] Uncaught Exceptions

test2.py > ...

```python
a = input()
b = 12
if a > 10:
    a -= 10
    b = b+a
    if a > 10:
        b = b+a
elif: a == 10:
    print(10)
else:
    print("a is not big enough.")

print(a+b)
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL 3: Python Debug Console

```
    main(ptvsdArgs)
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/__main__.py", line 432, in main
    run()
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/__main__.py", line 316, in run_file
    runpy.run_path(target, run_name='__main__')
  File "/anaconda3/lib/python3.7/runpy.py", line 261, in run_path
    code, fname = _get_code_from_file(run_name, path_name)
  File "/anaconda3/lib/python3.7/runpy.py", line 236, in _get_code_from_file
    code = compile(f.read(), fname, 'exec')
  File "/Users/ekapolc/Documents/python2/test2.py", line 8
    elif: a == 10:
        ^
SyntaxError: invalid syntax
bash-3.2$
```

ส่วนแสดงการทำงาน

test2.py — python2

DEBUG No Conf กดตรงนี้เพื่อเริ่ม

test2.py

VARIABLES

ส่วนแสดงตัวแปร

WATCH

CALL STACK

BREAKPOINTS

- [ ] Raised Exceptions
- [x] Uncaught Exceptions

test2.py > ...

```
a = input()
b = 12
if a > 10:
    a -= 10
    b = b+a
    if a > 10:
        b = b+a
elif: a == 10:
    print(10)
else:
    print("a is not big enough.")

print(a+b)
```

คลิ๊กด้านหน้าบรรทัดใดๆเพื่อสร้างเบรคพ้อยน์

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL 3: Python Debug Console

```
    main(ptvsdArgs)
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/__main__.py", line 432, in main
    run()
  File "/Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/lib/python/ptvsd/__main__.py", line 316, in run_file
    runpy.run_path(target, run_name='__main__')
  File "/anaconda3/lib/python3.7/runpy.py", line 261, in run_path
    code, fname = _get_code_from_file(run_name, path_name)
  File "/anaconda3/lib/python3.7/runpy.py", line 236, in _get_code_from_file
    code = compile(f.read(), fname, 'exec')
  File "/Users/ekapolc/Documents/python2/test2.py", line 8
    elif: a == 10:
        ^
SyntaxError: invalid syntax
bash-3.2$
```

ส่วนแสดงการทำงาน

test2.py — python2

DEBUG No Conf

VARIABLES

Locals

a: 10

b: 12

__builtins__: {'ArithmeticE...

__cached__: None

__doc__: None

__file__: '/Users/ekapolc/D...

__loader__: None

__name__: '__main__'

__package__: ''

__spec__: None

WATCH

CALL STACK PAUSED ON BREAKPO...

<module> test2.py 5:1

BREAKPOINTS

Raised Exceptions

Uncaught Exceptions

test2.py 5

test2.py

```
1  a =
2  b =
3  if a
4
5
6
7
8  elif
9
10 else
11
12
13 prin
```

ทำงานจนถึงเบรกพ้อยถัดไป

ทำงานบรรทัดถัดไป

ทำงานบรรทัดถัดไปในฟังก์ชั่น

ออกจากฟังก์ชั่น

เริ่มใหม่

หยุด

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 3: Python Debug Console

```
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60667 /Users/ekapolc/Documents/python2/test2.py
10
Terminated: 15
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60676 /Users/ekapolc/Documents/python2/test2.py
10
10
22
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60681 /Users/ekapolc/Documents/python2/test2.py
20
```

test2.py — python2

DEBUG No Conf

VARIABLES

Locals

a: 10

b: 12

__builtins__: {'ArithmeticE...

__cached__: None

__doc__: None

__file__: '/Users/ekapolc/D...

__loader__: None

__name__: '__main__'

__package__: ''

__spec__: None

WATCH

CALL STACK PAUSED ON BREAKPO...

<module> test2.py 5:1

BREAKPOINTS

Raised Exceptions

Uncaught Exceptions

test2.py 5

test2.py

test2.py > ...

```
2   b = 12
3   if a > 10:
4       a -= 10
9       print(10)
10  else:
11      print("a is not bi
12
13  print(a+b)
```

Remove Breakpoint
Edit Breakpoint...
Disable Breakpoint

Right click to edit break point

Expression breakpoint ทำงานเมื่อสิ่งนี้เป็นจริง เช่น x==3

Hitcount breakpoint ทำงานเมื่อทำงานบรรทัดนี้แล้วเท่ากับจำนวนครั้ง

Log message ปริ้นเมื่อเบรกพ้อยทำงาน ถ้าจะให้แสดงตัวแปร ใส่ค่าตัวแปรใน {} เช่น

Hi let's print {a}

ดูได้จาก Debug console

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 3: Python Debug Console

```
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60667 /Users/ekapolc/Documents/python2/test2.py
10
Terminated: 15
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60676 /Users/ekapolc/Documents/python2/test2.py
10
10
22
bash-3.2$ cd /Users/ekapolc/Documents/python2 ; env PYTHONIOENCODING=UTF-8 PYTHONUNBUFFERED=1 /anaconda3/bin/python /Users/ekapolc/.vscode/extensions/ms-python.python-2019.8.30787/pythonFiles/ptvsd_launcher.py --default --client --host localhost --port 60681 /Users/ekapolc/Documents/python2/test2.py
20
```

Workshop

Problem 1 : sum

Program received N as input, Return summation value of index 1 to N

index	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
value	1	2	6	4	15	12	7	8	18	30	11	24	13	14	150	16	17	36

N = 5 , output = 1+2+6+4+15 = 28
N = 15, output = 1+2+6+4+15+12+7+8+18+30+11+24+13+14+150 = 315

Example Input	Example
5	28
15	315

Problem 2 : coin

Given two type of coin, which have value A baht and B baht. Determine whether the target value T can be summed by the given types of coin.

6,8,10 -> A=6, B=8, target value=10; Coins 6 and 8 cannot sum up to 10

2,3,10 -> A=2, B=3, target value=10; Coins 2 and 3 can sum up to 10

If there are many possible answers, print any of them

Example Input	Example
6, 8, 10	IMPOSSIBLE
2, 3, 10	na: 2, nb: 2

Problem 2 : coin

find such a case that T = a*na + b*nb

Proposed algorithm

Check GCD(a,b)
https://www.khanacademy.org/computing/computer-science/cryptography/modarithmetic/a/the-euclidean-algorithm

```
loop find na
    if b | T - (a * na) then
        nb = ( T - (a*na) )/ b
```

Problem 3: anagram

A word is an **anagram** of another if the second is simply a re-arrangement of the first.

Example Input	Example
APPLE	['APPEL', 'APPLE', 'PEPLA']
PIE	['EPI', 'IPE', 'PEI', 'PIE']
noon	['NONO’, 'NOON']

Problem 3: anagram

Proposed algorithm: input list of all words

1. We gonna count number of alphabet as shown below,
   get_key(PIE) -> 0_0_0_0_1_0_0_0_1_0_0_0_0_0_0_1_0_0_0_0_0_0_0_0_0_0

0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

2.Then, we will store “PIE” into the dictionary, using our precompute key

dict[get_key(PIE)].append(“PIE”)

3.In the end, dictionary should contains all words with the same alphabet count

dict[get_key(PIE)] = ['EPI', 'IPE', 'PEI', 'PIE']

IEP
PI

Mydict = {}

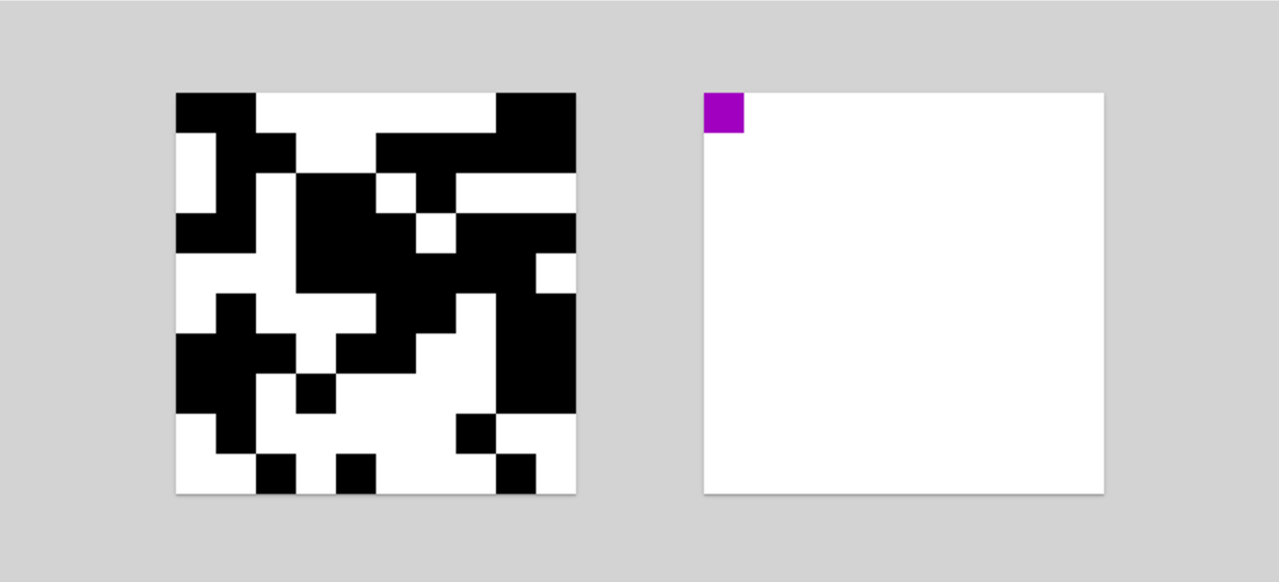
Mydict = { P1I1E1:[PIE, IEP], P1I1:[PI] }

IP -> P1I1

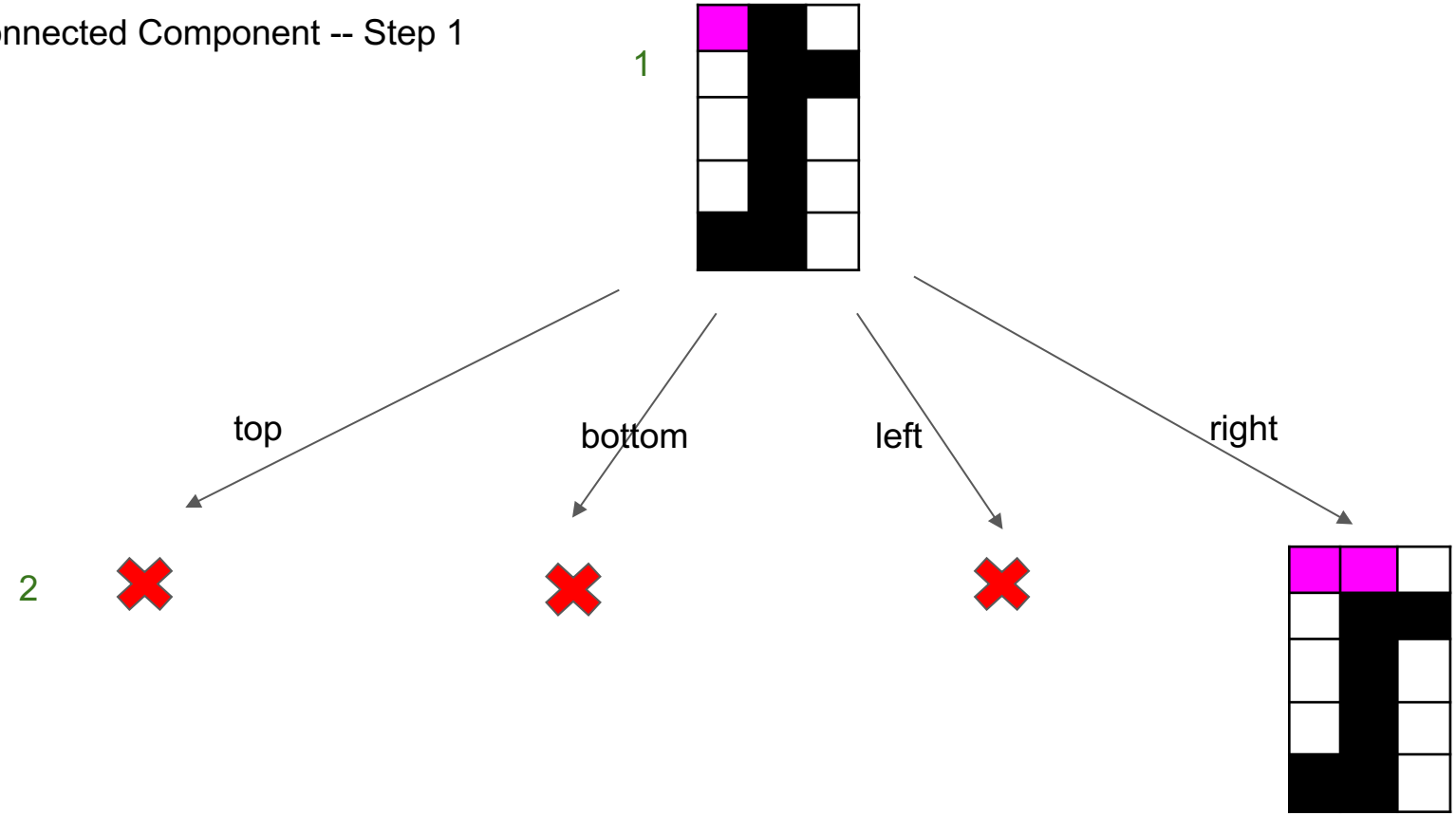
Mydict[P1I1] -> [PI]

Problem 4: Connected component

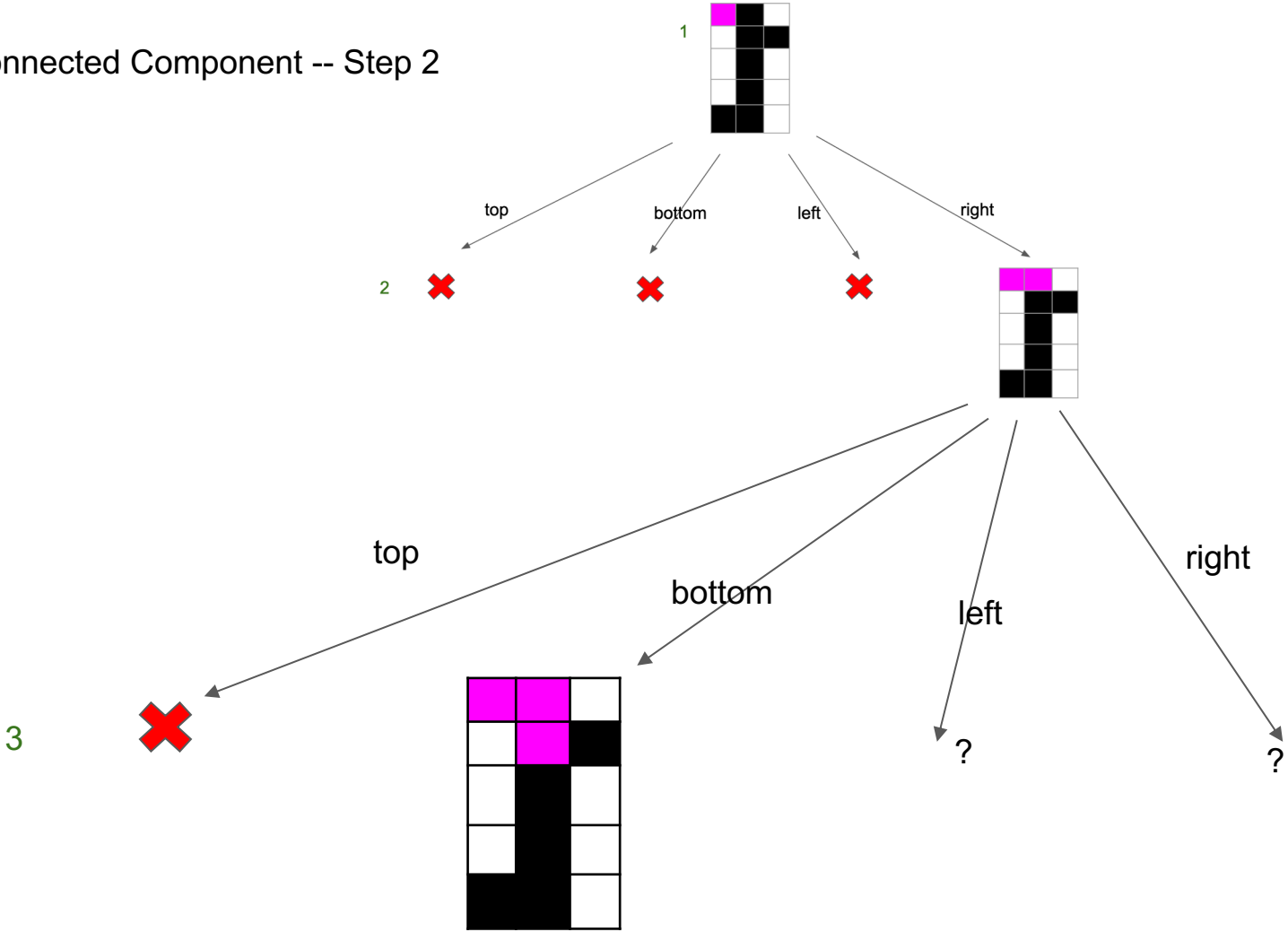
Visualized coloring algorithm



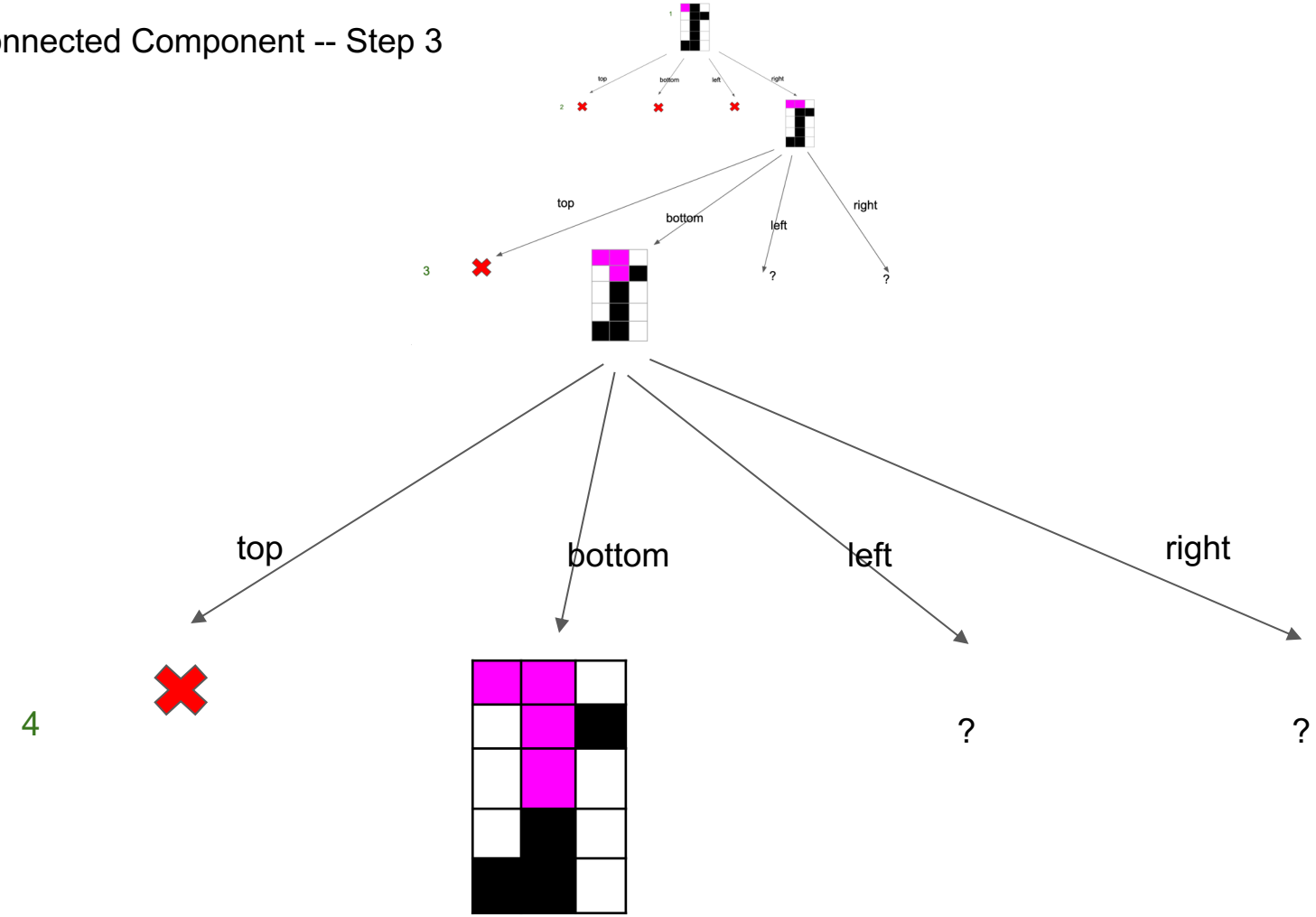
Connected Component -- Step 1



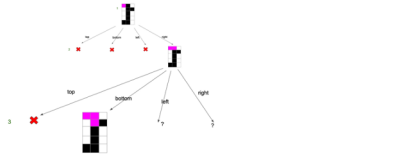
Connected Component -- Step 2

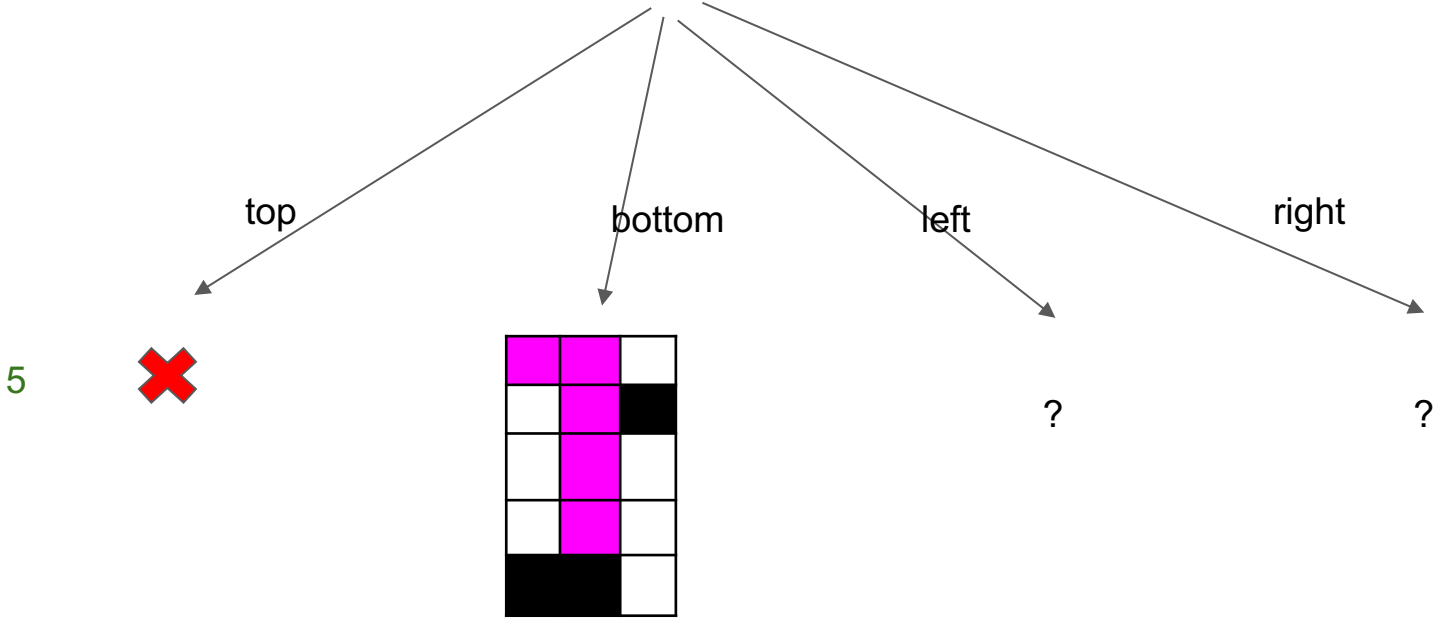


Connected Component -- Step 3

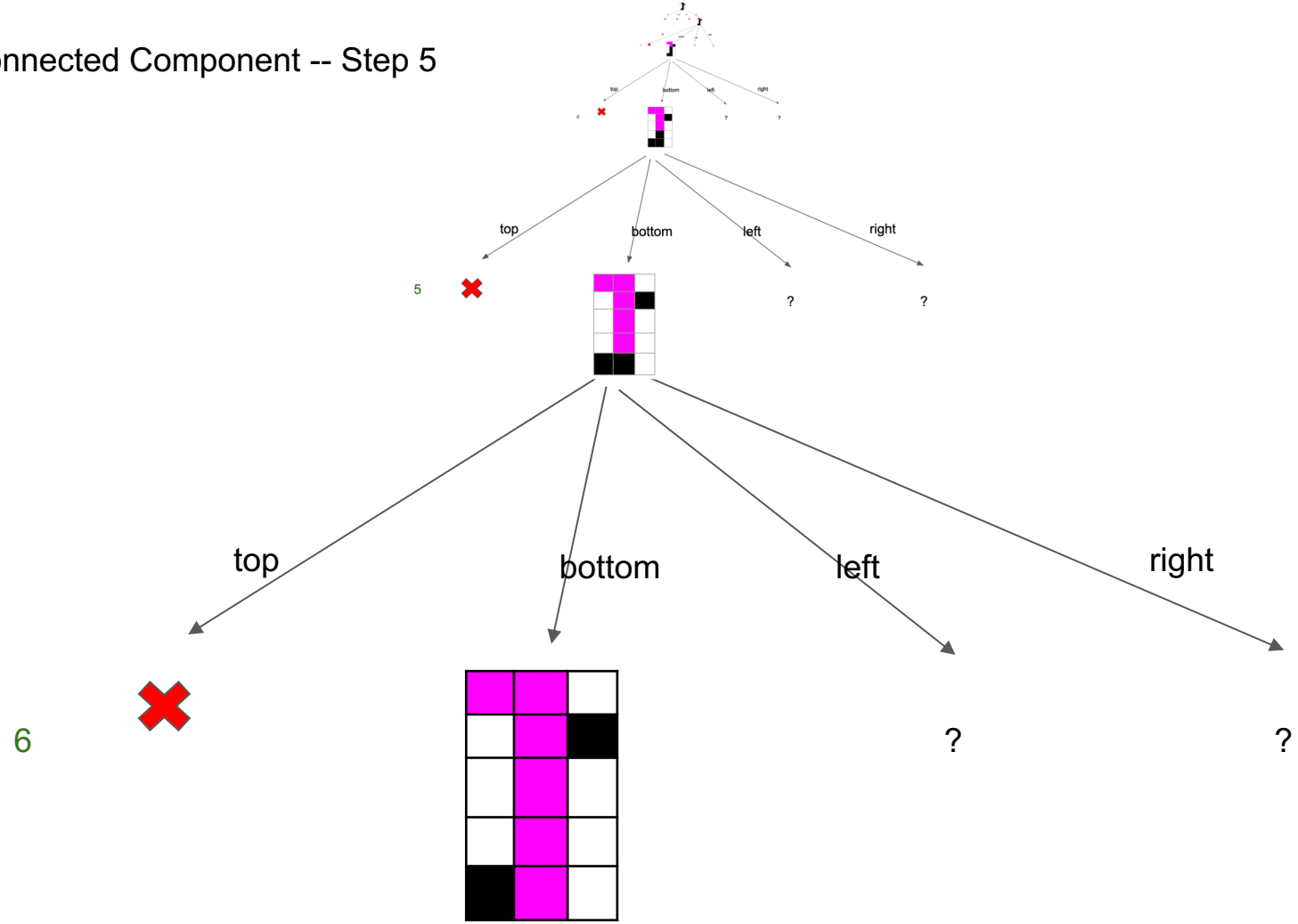


Connected Component -- Step 4

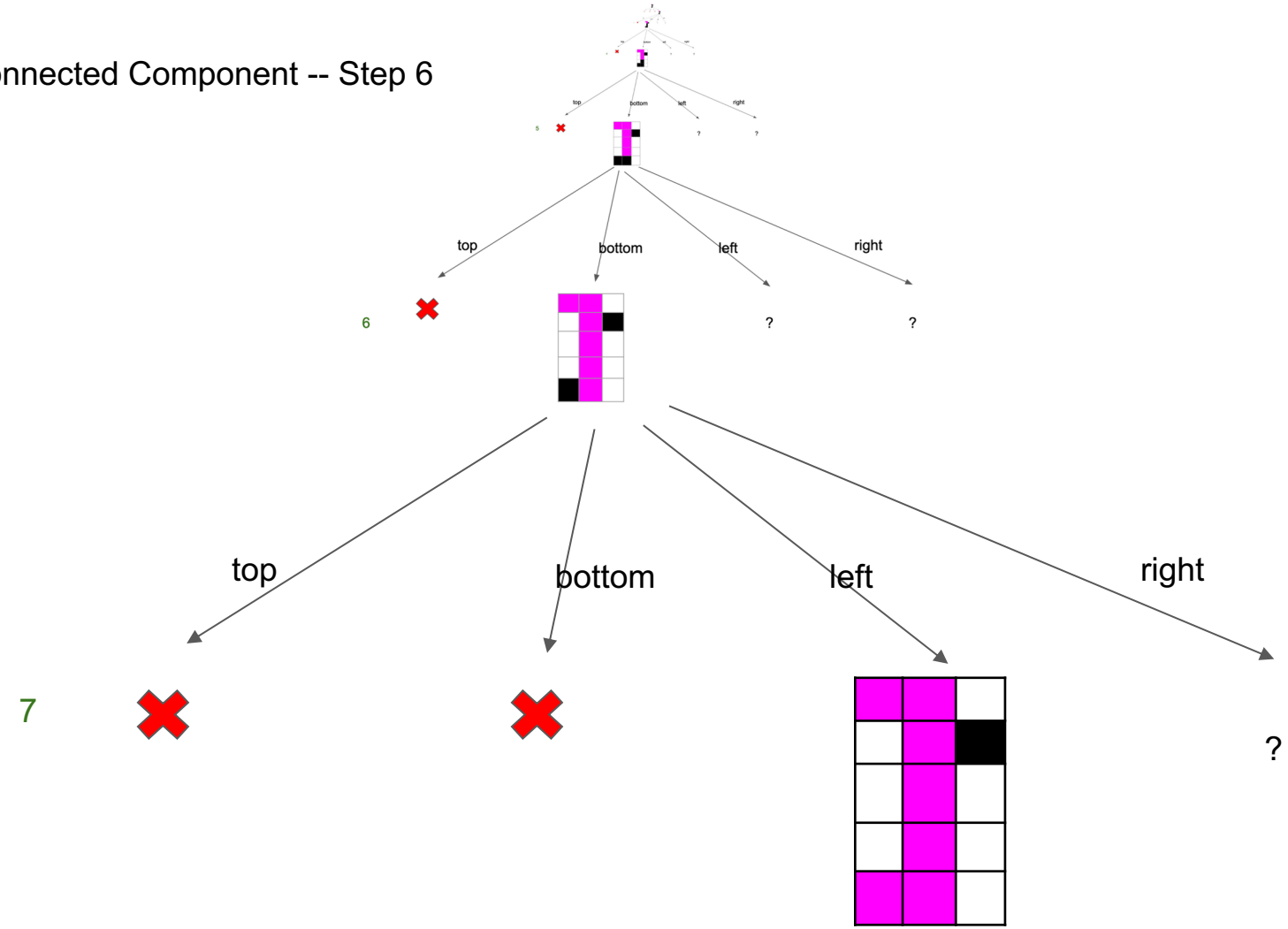




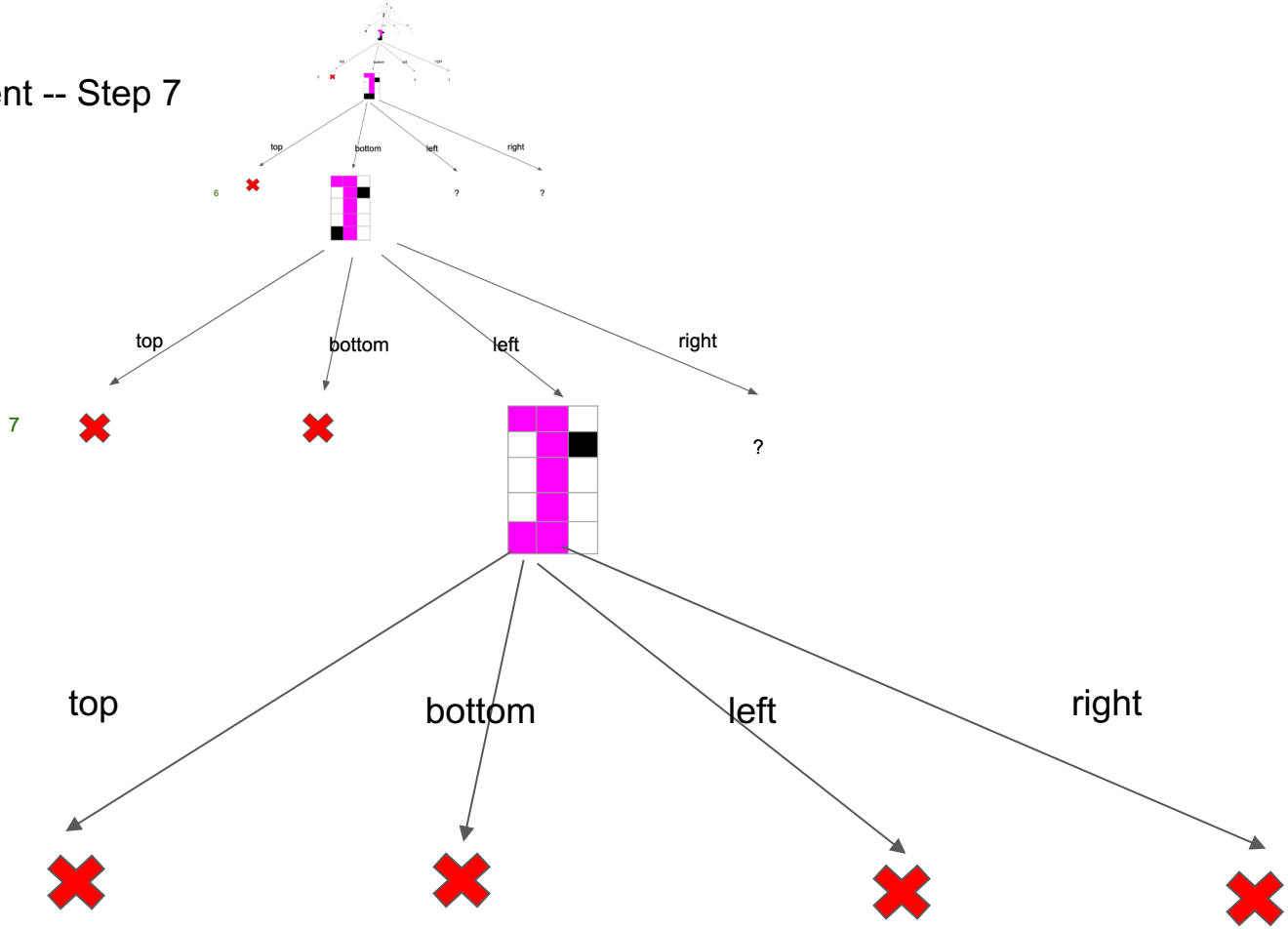
Connected Component -- Step 5



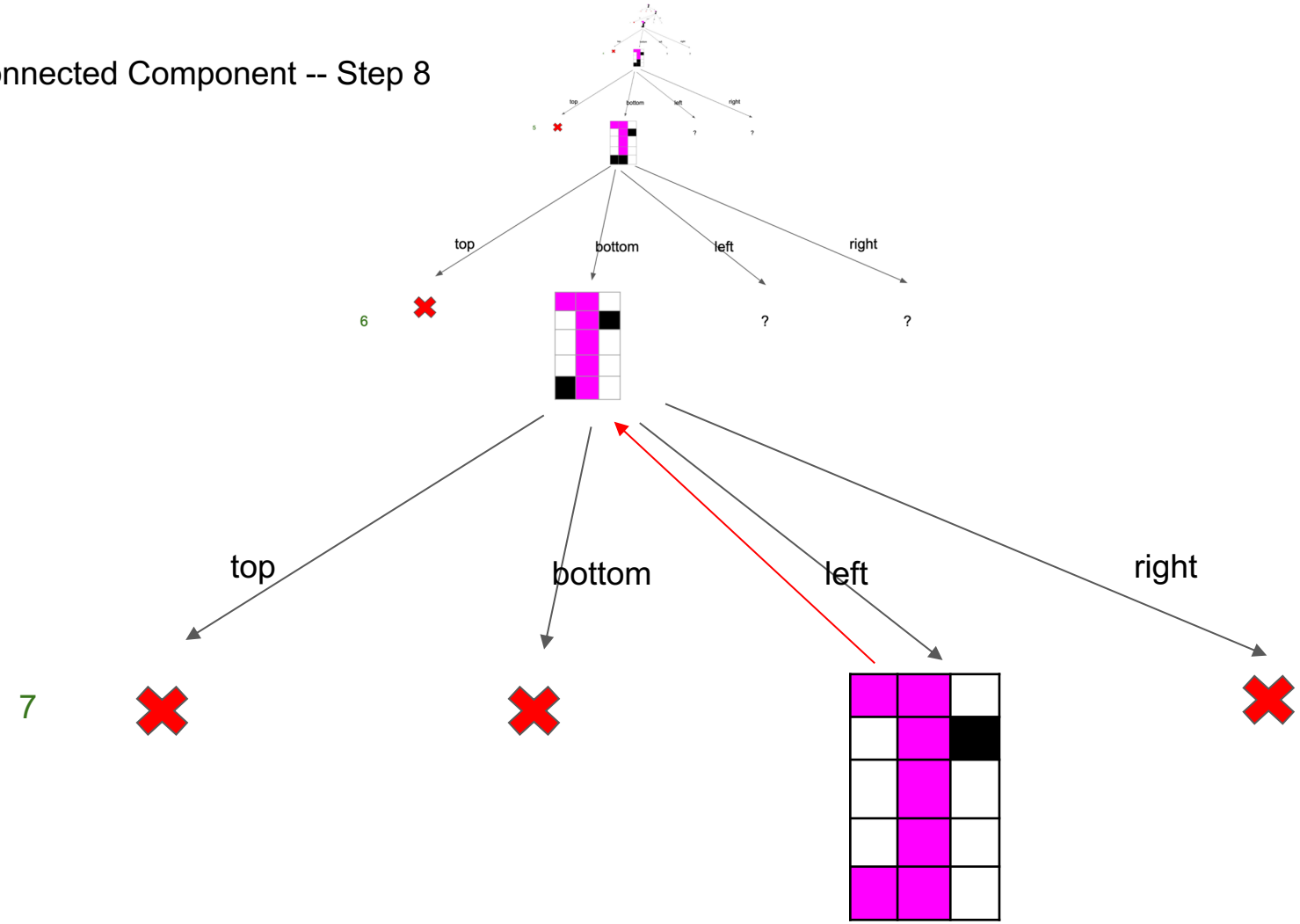
Connected Component -- Step 6



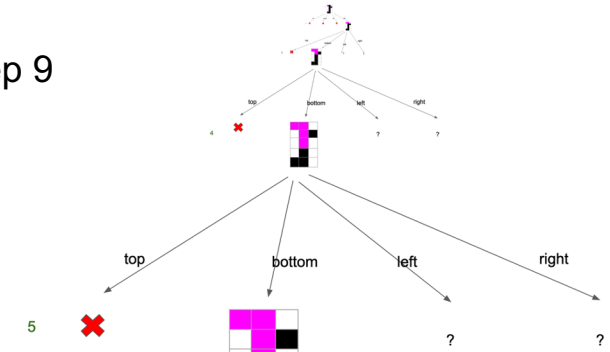
Connected Component -- Step 7

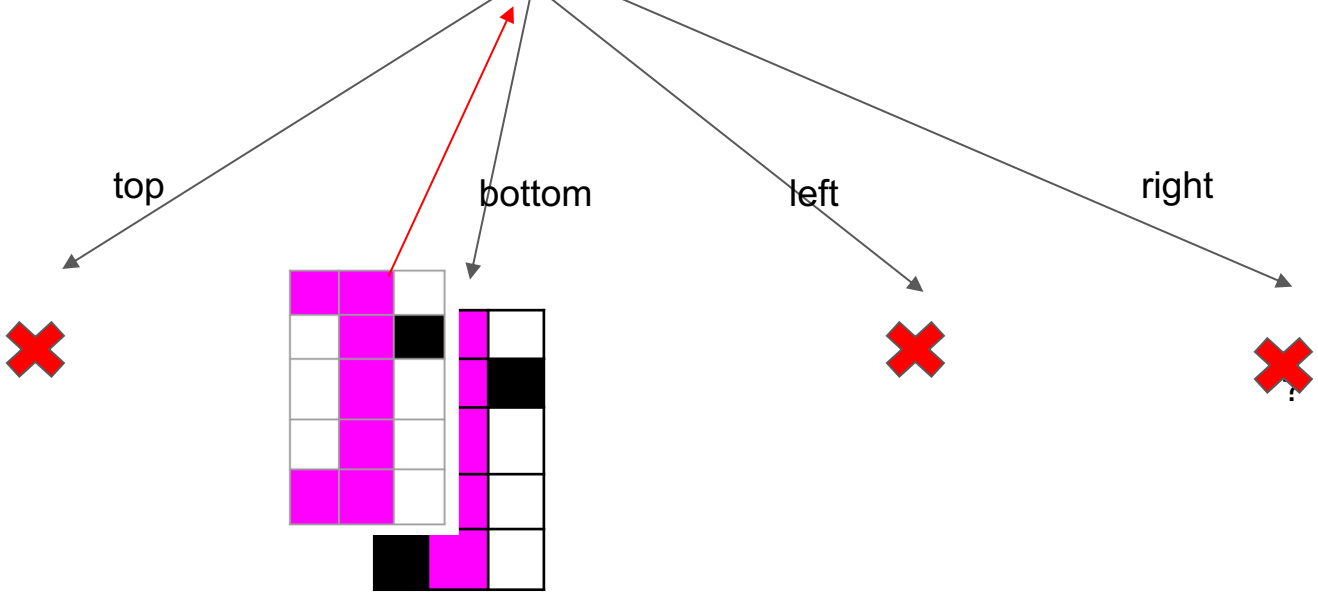


Connected Component -- Step 8

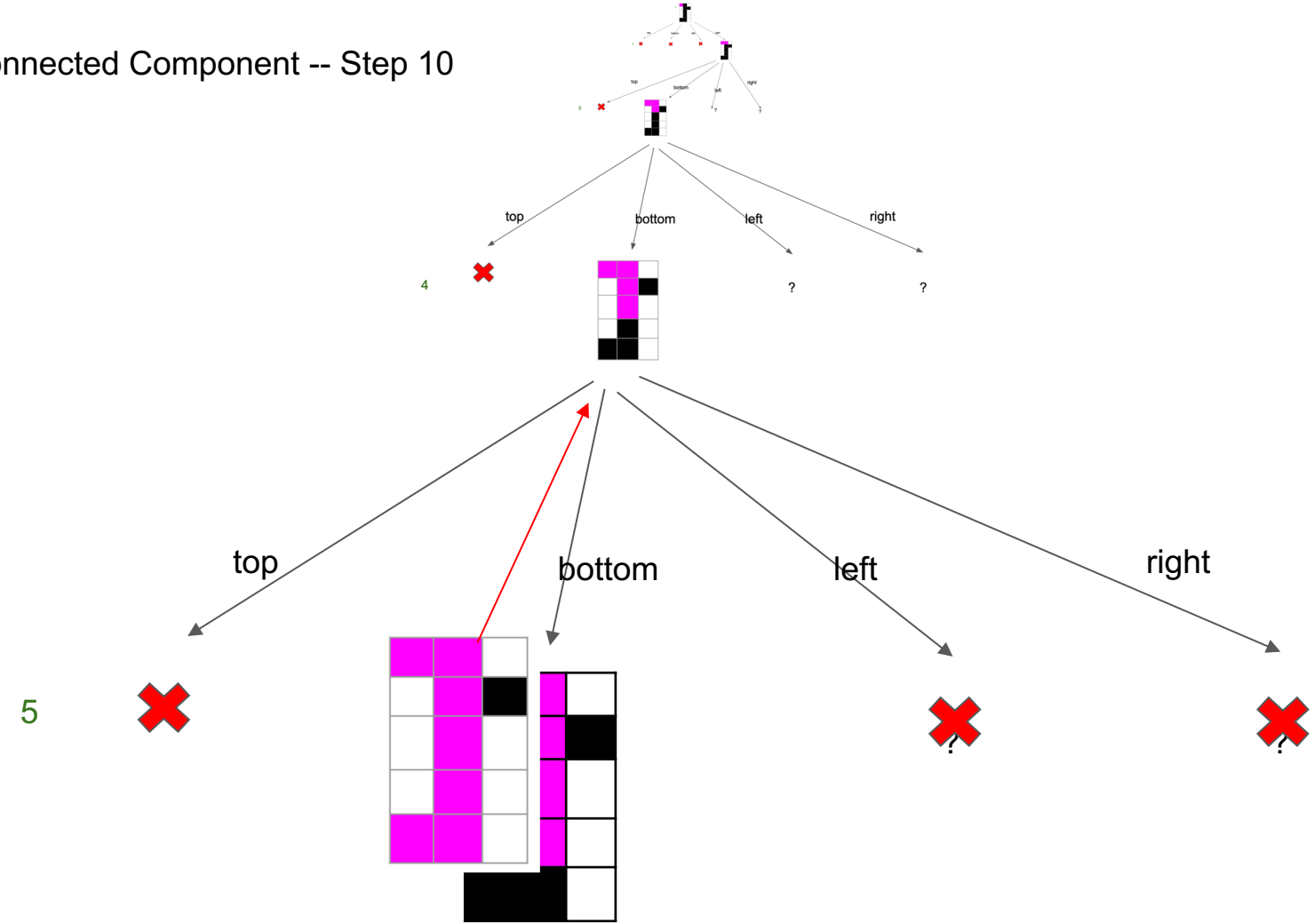


Connected Component -- Step 9

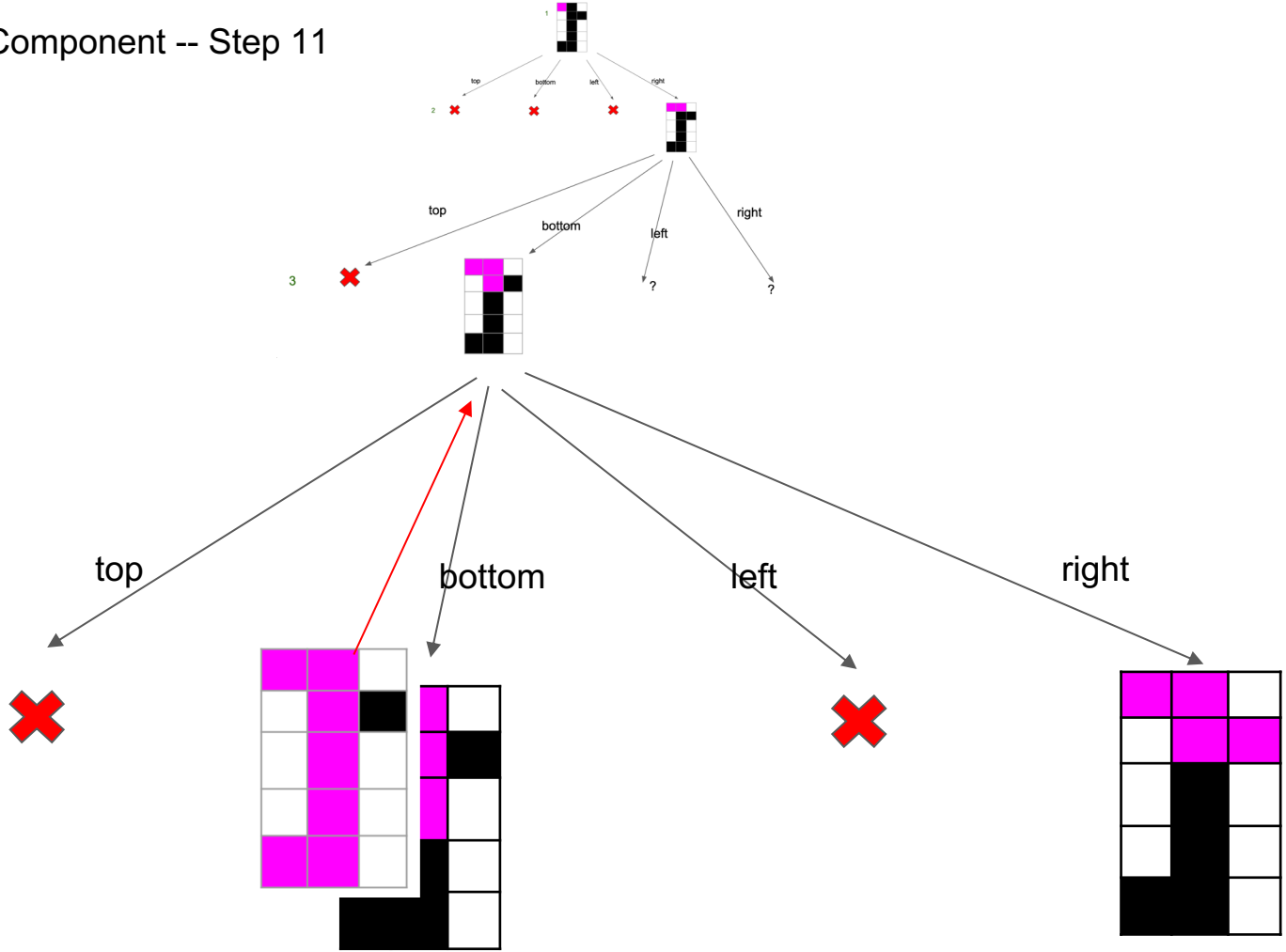




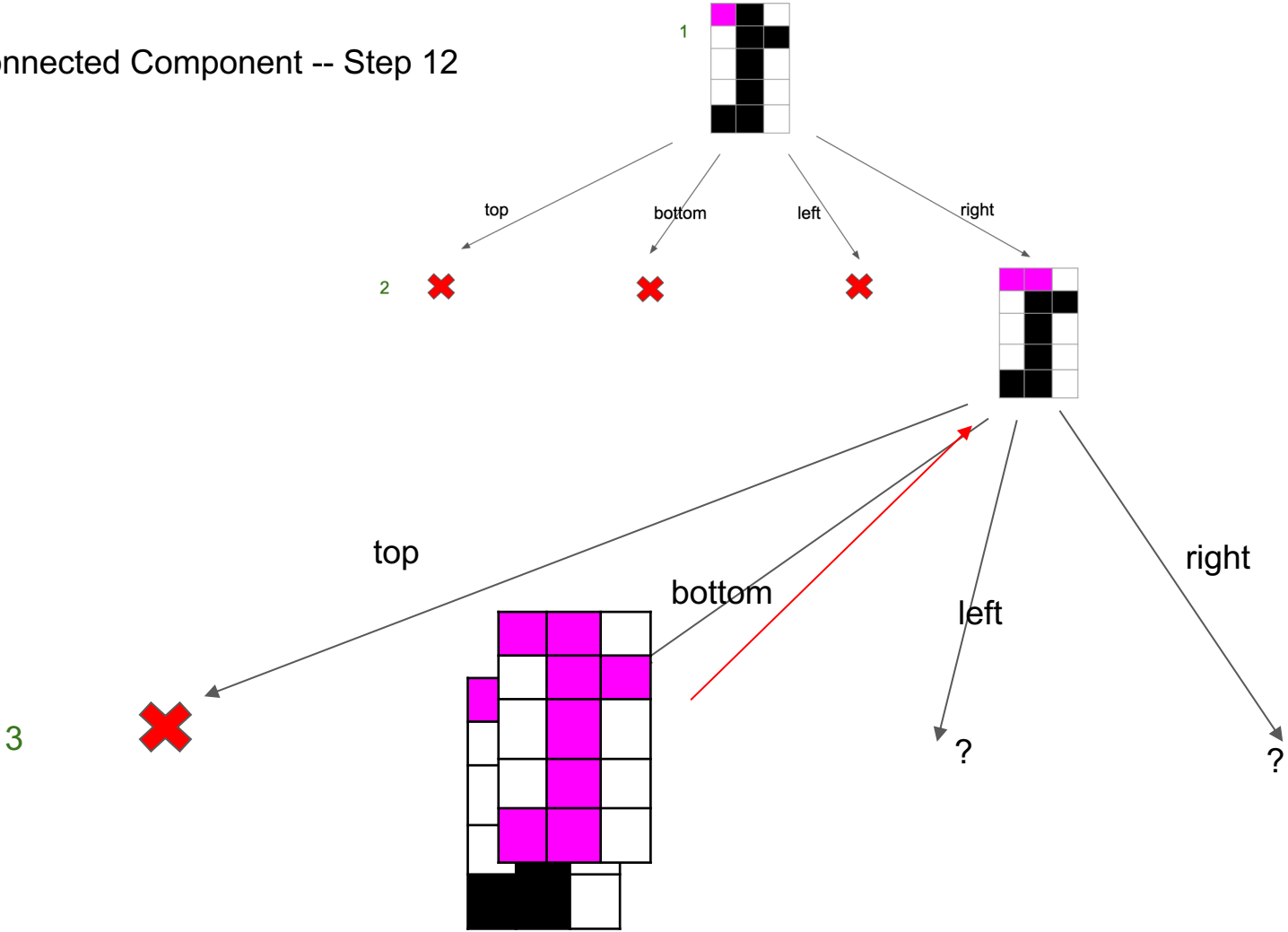
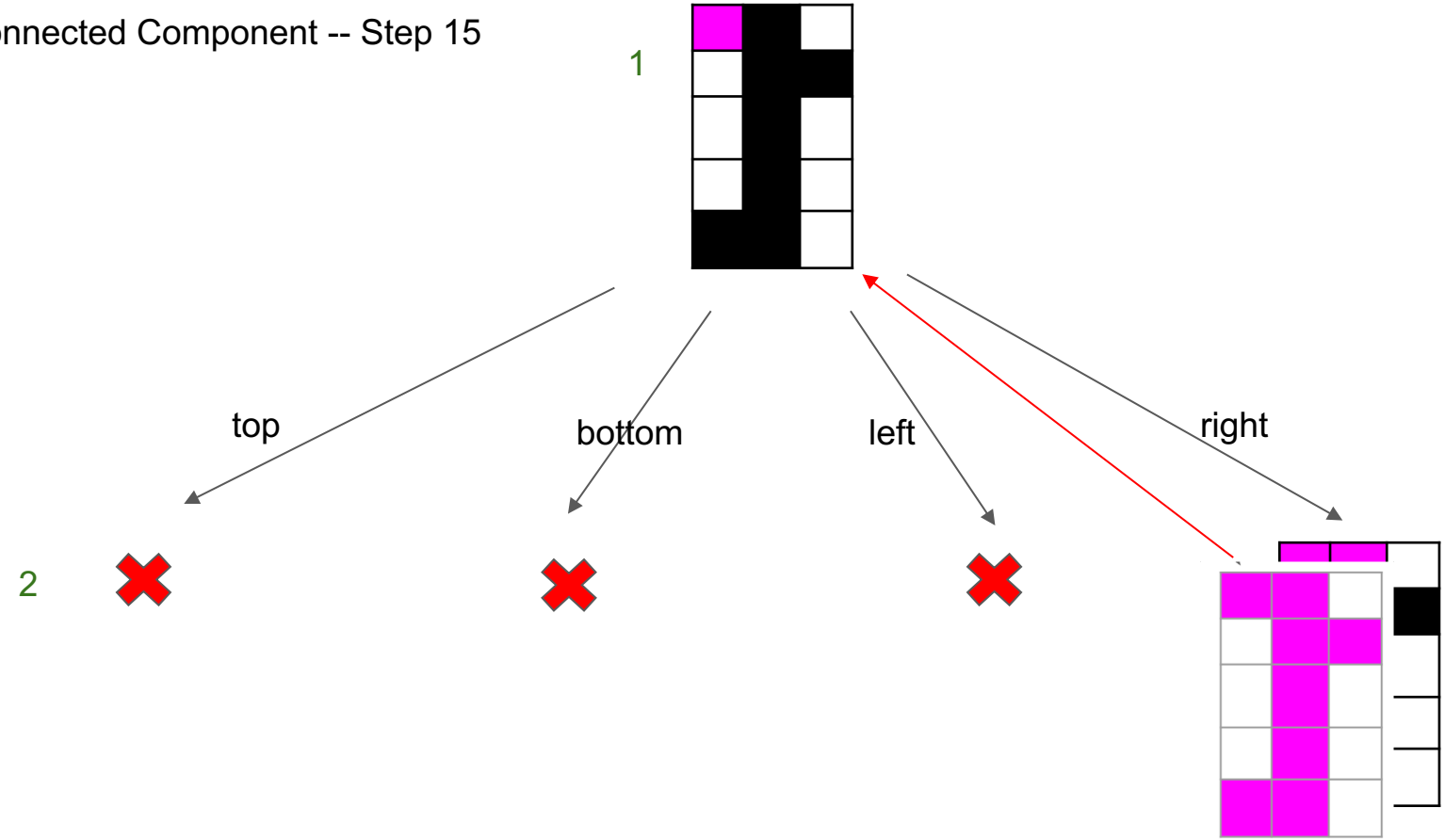
Connected Component -- Step 10



Connected Component -- Step 11



Connected Component -- Step 12



Connected Component -- Step 15